

SQL & Databases

Talk to databases · Equigrow short notes

Basics

```
SELECT name, cgpa FROM students
WHERE branch = 'CSE' AND cgpa >= 8
ORDER BY cgpa DESC
LIMIT 10;
```

Order of writing: SELECT → FROM → WHERE → GROUP BY → HAVING → ORDER BY → LIMIT.

Aggregates & grouping

```
SELECT branch, COUNT(*) AS students, AVG(cgpa) AS avg_cgpa
FROM students
GROUP BY branch
HAVING COUNT(*) > 50;
```

WHERE filters rows before grouping; **HAVING** filters groups after.

Joins

```
SELECT s.name, c.title
FROM students s
JOIN enrolments e ON e.student_id = s.id
JOIN courses c ON c.id = e.course_id;
```

Join	Returns
INNER JOIN	only rows that match in both tables
LEFT JOIN	all rows from the left table, NULL where no match
RIGHT JOIN	all rows from the right table
FULL JOIN	all rows from both

Changing data & tables

```
INSERT INTO students (name, branch) VALUES ('Asha', 'CSE');
UPDATE students SET cgpa = 8.9 WHERE id = 7;
DELETE FROM students WHERE id = 7;
CREATE TABLE courses (id INT PRIMARY KEY, title VARCHAR(100) NOT NULL);
```

Key concepts

- **Primary key:** unique id for each row. **Foreign key:** links to another table's primary key.
- **Index:** makes searches faster, slows writes slightly.
- **Normalisation:** split data so each fact is stored once (1NF, 2NF, 3NF).
- **ACID:** Atomicity, Consistency, Isolation, Durability: what makes transactions safe.